SHRAVAN M KUMAR

Atlanta, Georgia | (609) 480-5988 | skumar803@gatech.edu| linkedin.com/in/shravanmkumar
Link to Portfolio: *Link Here*

EDUCATION

Georgia Institute of Technology, Atlanta, GA — August 2026 – May 2029
BS in Mechanical Engineering
Relevant Coursework: Heat Transfer, Fluid Dynamics, Thermodynamics, Mechanics (Full)
GPA: Available to send by December 2026

Rutgers University, New Brunswick, NJ — August 2025 – May 2026
GPA: 3.97/4.00

EXPERIENCE

Propulsion Engineer, *Propulsive Landers @ Georgia Tech (GTPL)* — Atlanta, GA — August 2026 – Present
- Contributed propulsion sub team in preparing [component/system] for the team's hybrid rocket engine hotfire campaign, [your specific task — e.g., instrumenting the test stand, machining injector components, running propellant flow calcs]
- Assisted with ground testing and characterization of [oxidizer feed system / injector / motor casing] ahead of tethered hover trials, logging [chamber pressure / thrust / flow rate] data during static fires
- [GNC/avionics-adjacent bullet if you touch that side — e.g., supported integration between propulsion and guidance systems for the vertical takeoff-and-landing test vehicle]

Propulsion Engineer, *Georgia Tech Experimental Rocketry Club (GTXR)* - Atlanta, GA — August 2026 – Present
- Contributed to the propulsion sub team in fabricating and assembling solid motor components, including nozzle assemblies and motor casings for 6" and 8" motors
- Assisted with static fire test setup and post-test data collection (chamber pressure, thrust curves)
- Trained on propellant handling and casting procedures under senior-member supervision

Web Development Intern, *Cardinal Integrated Technologies* - Skillman, NJ — April 2024 – June 2024
- Built a customer-facing store in React.js with hooks and Fetch API, collaborating with an interdisciplinary team to refine usability

Competitive Analysis Intern, *Ziplyne Technologies* - Skillman, NJ — July 2023 – October 2023
- Conducted systematic analysis of 138 competing products, delivering 13 presentations summarizing key insights and recommendations to 10 product and strategy stakeholders

PROJECTS (PORTFOLIO LINK)

5- DOF Robot Arm — June 2026 – August 2026
- Implemented real-time position control in MATLAB/Simulink using OpenCV based object detection to drive servo actuation to pick up identified object using 2 pose vector analysis in python to pick up objects in 3d with one camera
- Designed and modeled a 5-DOF Robotic Arm in SolidWorks, then imported into Simscape Multibody to run Inverse Kinematics calculations, running local hosts between OpenCV python, MATLAB, and Serial Bus Driver
- Investigated Model Predictive Control Applications and the gap between simulation and real system behavior to inform controller design

Mitsubishi 6G72 CAD Assembly — January 2026 – May 2026
- Designed and assembled a fully functional V6 engine in SolidWorks, modeling components to accurate dimensions including pistons, crankshaft, camshaft, rockers, and cylinder block with accurate geometric constraints
- Validated mechanical motion using SolidWorks Motion Study simulation

SKILLS

- Ansys, FEA, Simulink (MATLAB), SolidWorks, Autodesk Inventor (Design, Assembly, 2D Drawings), Python, React.js, HTML, CSS